Boston Bus Equity

October Project Check-In Report, CS506 / Spark!
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Key terms

Term	What it means
Timepoint	A bus stop where the MBTA checks whether a bus is keeping to its schedule. A route has only a few of these.
Crossing	One bus passing one timepoint. A single trip creates several crossings.
On time	A crossing that passes the MBTA's reliability test. The on-time rate is the share of crossings that pass.
Target routes	15 priority routes from the Livable Streets report: 14, 15, 17, 22, 23, 24, 26, 28, 29, 31, 33, 42, 44, 45, 111.

Data Processing

What are our data sources, and how did we collect them?

All of our data is public, and our notebook downloads it automatically.

Data	Where it came from	How we got it	What we used it for
Bus arrival and departure times, 2020, 2024	MBTA	Downloaded files	Reliability, wait, trip time
Bus ridership by route, Fall 2016-2025	MBTA	Downloaded files	Number of riders on each route
Rider survey, 2025 (latest public)	MBTA	API	Riders' race and income
Rider survey, 2015-17	Planning agency (CTPS)	Downloaded file	Comparison with earlier riders
Bus stop locations	MBTA	Downloaded file	Placing each stop on the map
Census income and race, 2020-24	Census Reporter	API	Who lives near each stop
Census tract boundaries	US Census	API	Matching stops to tracts
Service Delivery Policy, Fall 2024	MBTA	Read the policy	The definition of on time

What cleaning and processing did we do, and why?

What we did	Why
Compared Jan-Feb 2020 with Jan-Feb 2024	These are the last pre-COVID months available, and using the same months removes seasonal effects.
Kept weekdays only, and stopped 2024 at November	Weekend schedules differ, and the network changed in December 2024.
Matched route names across files (SL1 = 741)	Files name routes differently, so busy routes would otherwise be lost.
Left out 1.7 million blank arrival times	A blank can be a canceled trip or a data gap. Counting them all as not on time lowers the 2024 on-time rate from 66.3% to 61.4%.
Removed 2,025 extreme delays (over 30 min early or 2 hours late)	These are likely tracking errors.
Matched bus stops to census tracts	This links bus performance to neighborhood income and race.

Check: our Fall 2018 ridership is within 14% of the Livable Streets figures on all 15 target routes.

Why did we choose these features?

Feature	Why
On-time rate	This is the MBTA's own reliability standard.
Average delay	It shows how late buses run, which the on-time test can miss on frequent routes.
Expected wait	It estimates how long riders wait at the stop.
Trip time	It shows how long trips take and how predictable they are.
Weekday boardings	They show how many people each route affects.
Race and income	They show who is affected, both for riders (survey) and for neighborhoods (census).

Hypotheses Tested

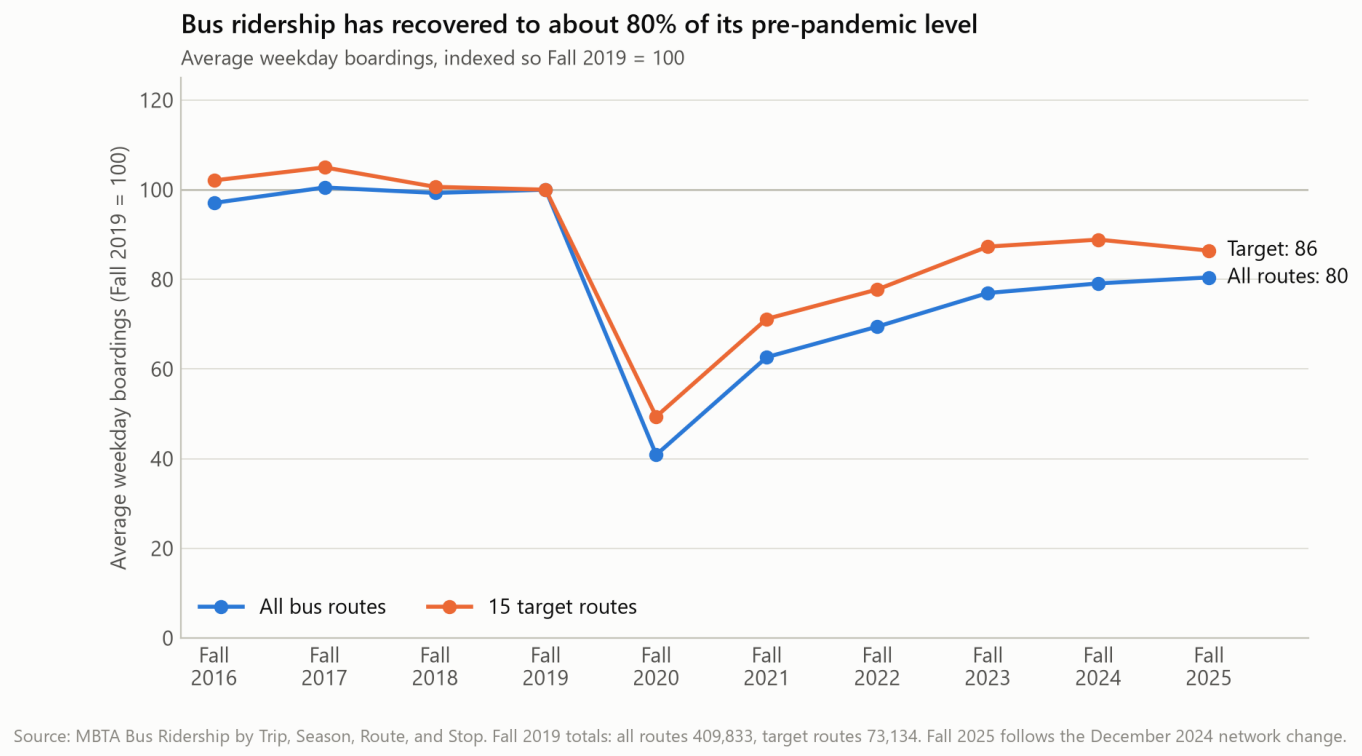
"Yes" means the data back up the claim, but it does not prove what caused it.

#	What we expected	True?	What we found	Figures
H1	Fewer people ride the bus than before the pandemic, and the drop differs by route	**Yes**	Weekday riders are at 79% of their Fall 2019 level, and the typical route lost 20%.	1, 2
H2	Bus trips take longer and are less predictable than scheduled	**Partly**	A typical trip takes about as long as scheduled, but fast and slow trips on a route differ by about 16 minutes.	3
H3	Riders wait longer than the schedule suggests	**Yes**	On frequent routes, the wait is 8.6 minutes, not the 6.9 the schedule suggests.	4
H4	Some routes are much more reliable than others	**Yes**	Routes range from 40% to 80% on time.	5
H5	The 15 target routes perform worse than other routes	**Partly**	They are on time about as often, but run about 1.4 minutes later on average.	5
H6	Minority and low-income communities get worse service	**Partly**	On-time rates are about the same across neighborhoods, but buses run somewhat later in minority and lower-income areas.	7, 8, 9
H7	A small number of routes cause most of the lost time	**Weakly**	The 15 routes losing the most time account for 45% of lost hours, but they also carry 38% of riders.	10
H8	Buses were less reliable in 2024 than before the pandemic	**Yes**	From early 2020 to early 2024, the on-time rate fell from 73% to 69%, and average delay rose from 2.3 to 3.9 minutes.	6

Data Visualizations

Figure 1. Ridership over time

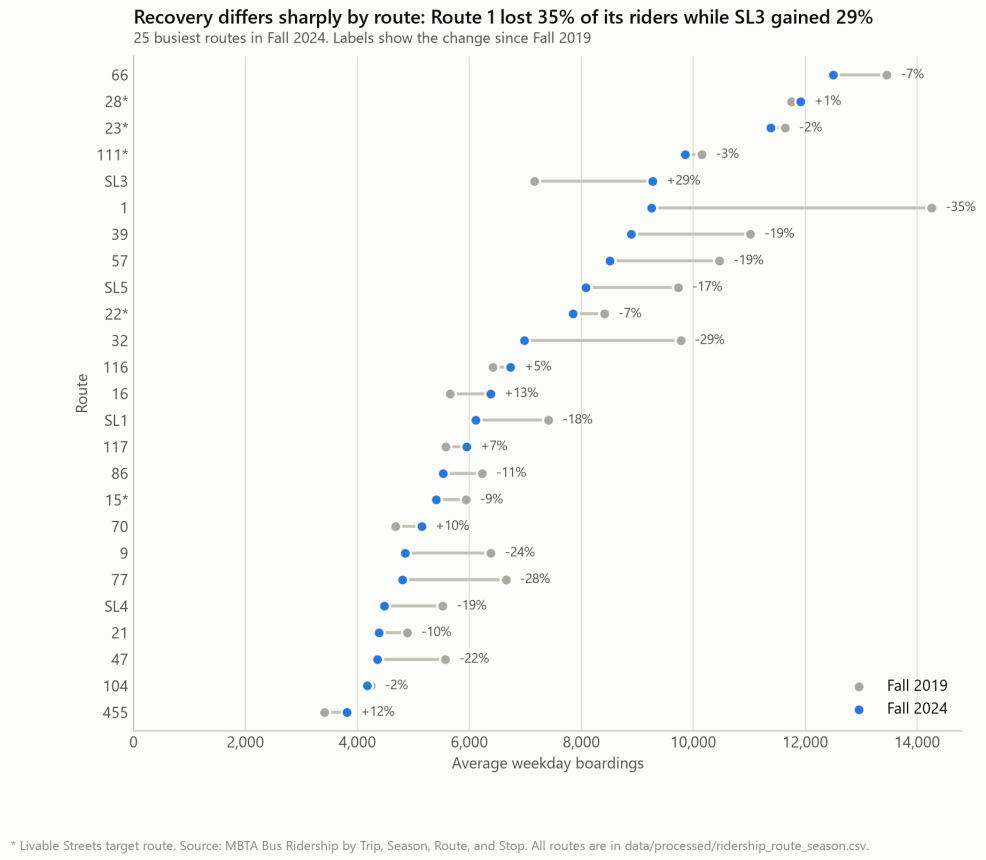
Verdict: True: bus ridership has not fully come back, with weekday riders at about 80% of their Fall 2019 level.



Source: MBTA Bus Ridership by Trip, Season, Route, and Stop. Fall 2019 totals: all routes 409,833, target routes 73,134. Fall 2025 follows the December 2024 network change.

Figure 2. Ridership by route

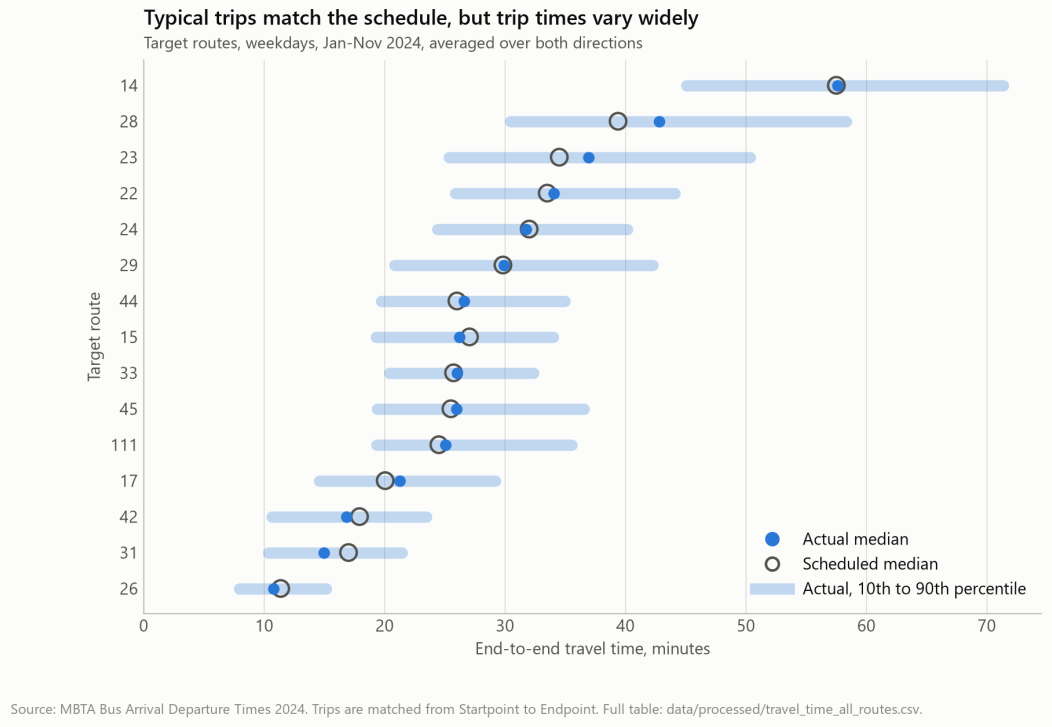
Verdict: True: the recovery is uneven, since the typical route lost 20% of its riders and only 18 of 127 routes gained riders.



* Livable Streets target route. Source: MBTA Bus Ridership by Trip, Season, Route, and Stop. All routes are in data/processed/ridership_route_season.csv.

Figure 3. End-to-end travel time

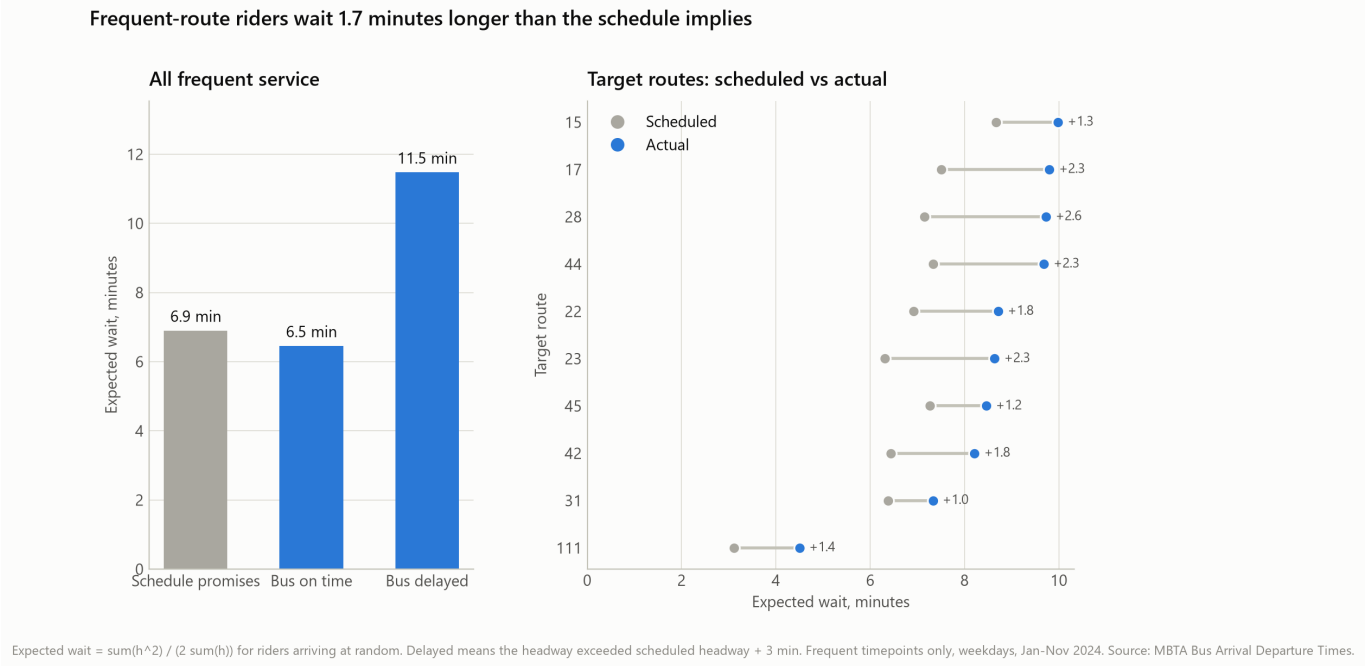
Verdict: Partly true: a typical trip takes about as long as scheduled, but fast and slow trips on a route differ by about 16 minutes.



Source: MBTA Bus Arrival Departure Times 2024. Trips are matched from Startpoint to Endpoint. Full table: data/processed/travel_time_all_routes.csv.

Figure 4. Rider wait time

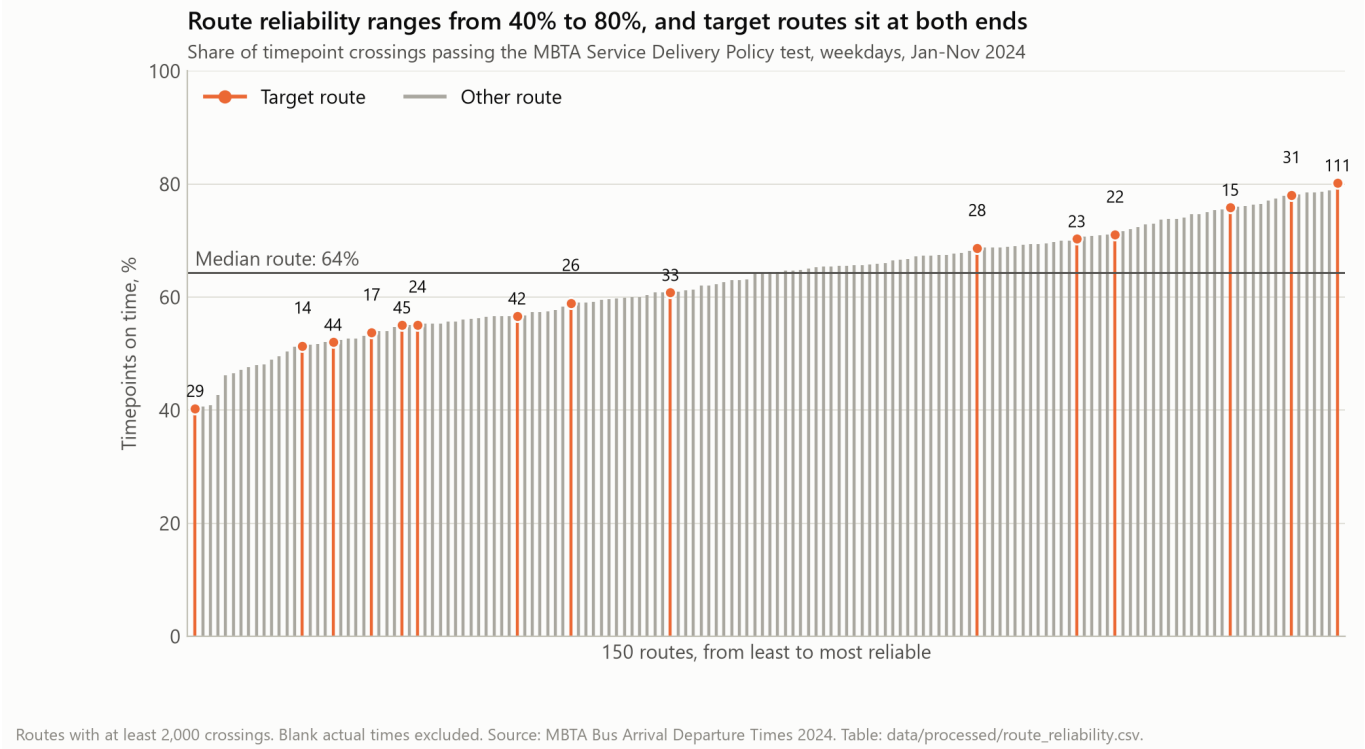
Verdict: True: riders on frequent routes wait about 8.6 minutes instead of the 6.9 the schedule suggests, and about 11.5 when a bus is delayed.



Expected wait = sum(h^2) / (2 sum(h)) for riders arriving at random. Delayed means the headway exceeded scheduled headway + 3 min. Frequent timepoints only, weekdays, Jan-Nov 2024. Source: MBTA Bus Arrival Departure Times.

Figure 5. Reliability by route

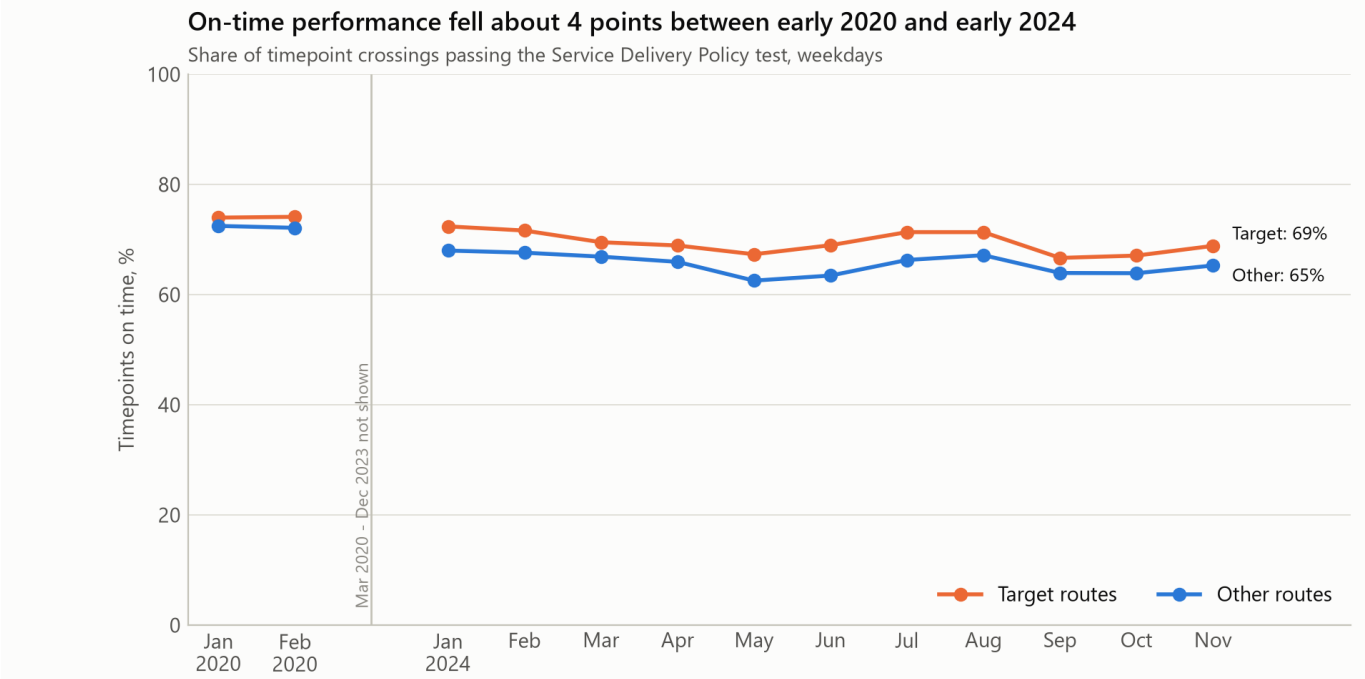
Verdict: True for routes overall, partly true for target routes: routes range from 40% to 80% on time, and target routes are on time about as often as others but run about 1.4 minutes later.



Routes with at least 2,000 crossings. Blank actual times excluded. Source: MBTA Bus Arrival Departure Times 2024. Table: data/processed/route_reliability.csv.

Figure 6. Reliability over time

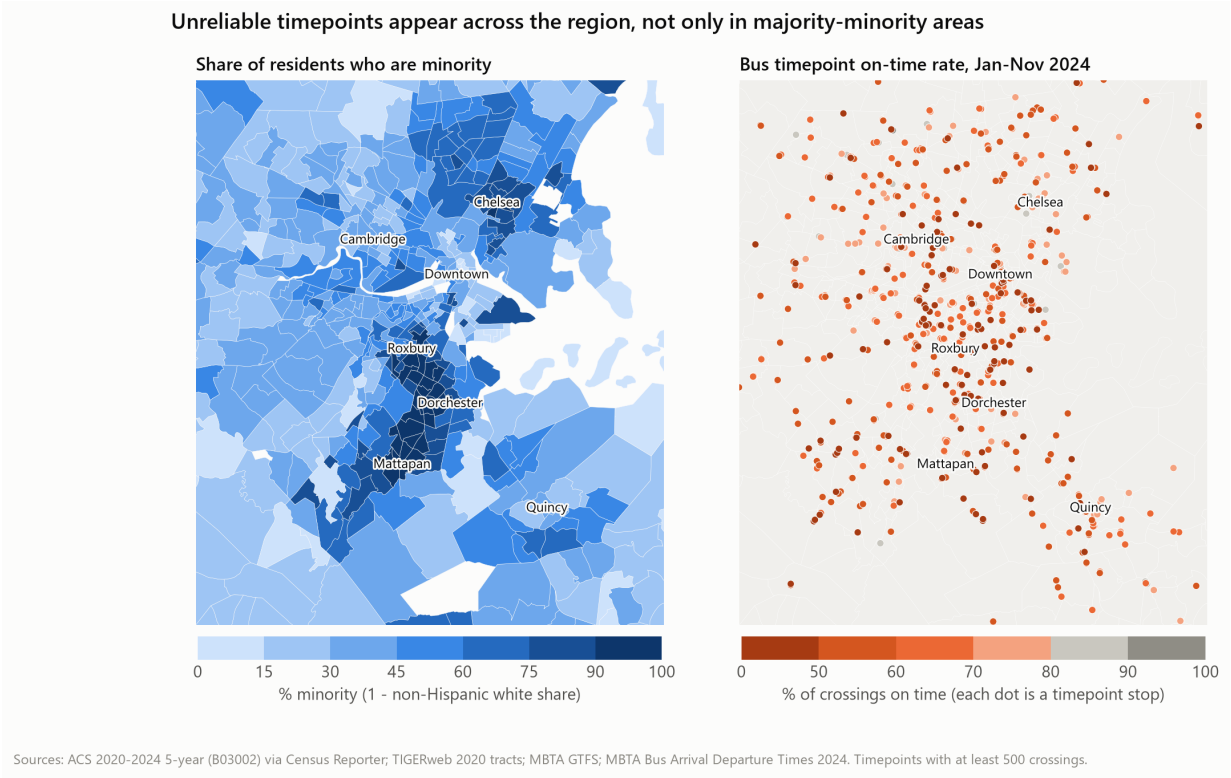
Verdict: True: between early 2020 and early 2024, the on-time rate fell from 73% to 69%, and average delay rose from 2.3 to 3.9 minutes.



Pooled by timepoint crossing. Source: MBTA Bus Arrival Departure Times 2020 and 2024. 2018-2019 are not publicly downloadable.

Figure 7. Reliability and neighborhoods

Verdict: No clear pattern: stops with poor on-time rates appear across the whole region, not only in mostly minority neighborhoods.



Sources: ACS 2020-2024 5-year (B03002) via Census Reporter; TIGERweb 2020 tracts; MBTA GTFS; MBTA Bus Arrival Departure Times 2024. Timepoints with at least 500 crossings.

Figure 8. Rider demographics by route

Verdict: No clear link: routes with more minority or low-income riders are not clearly less on time, but this check covers only 56% of riders.

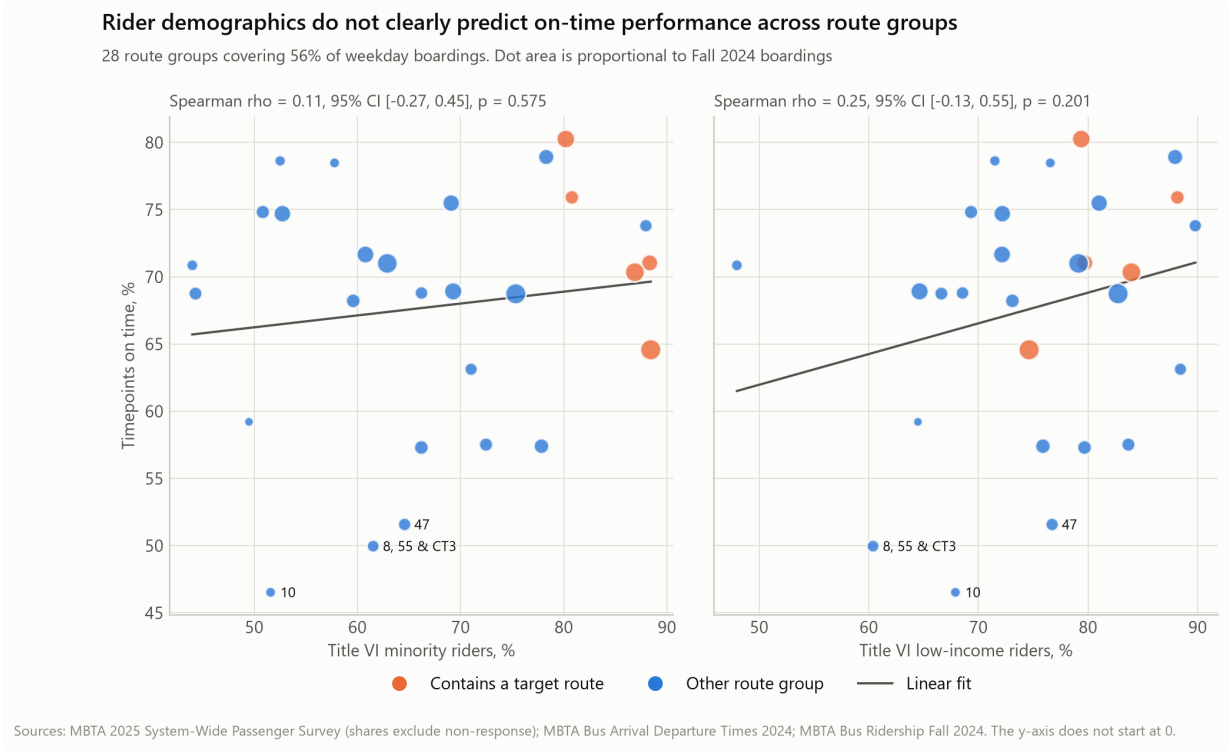


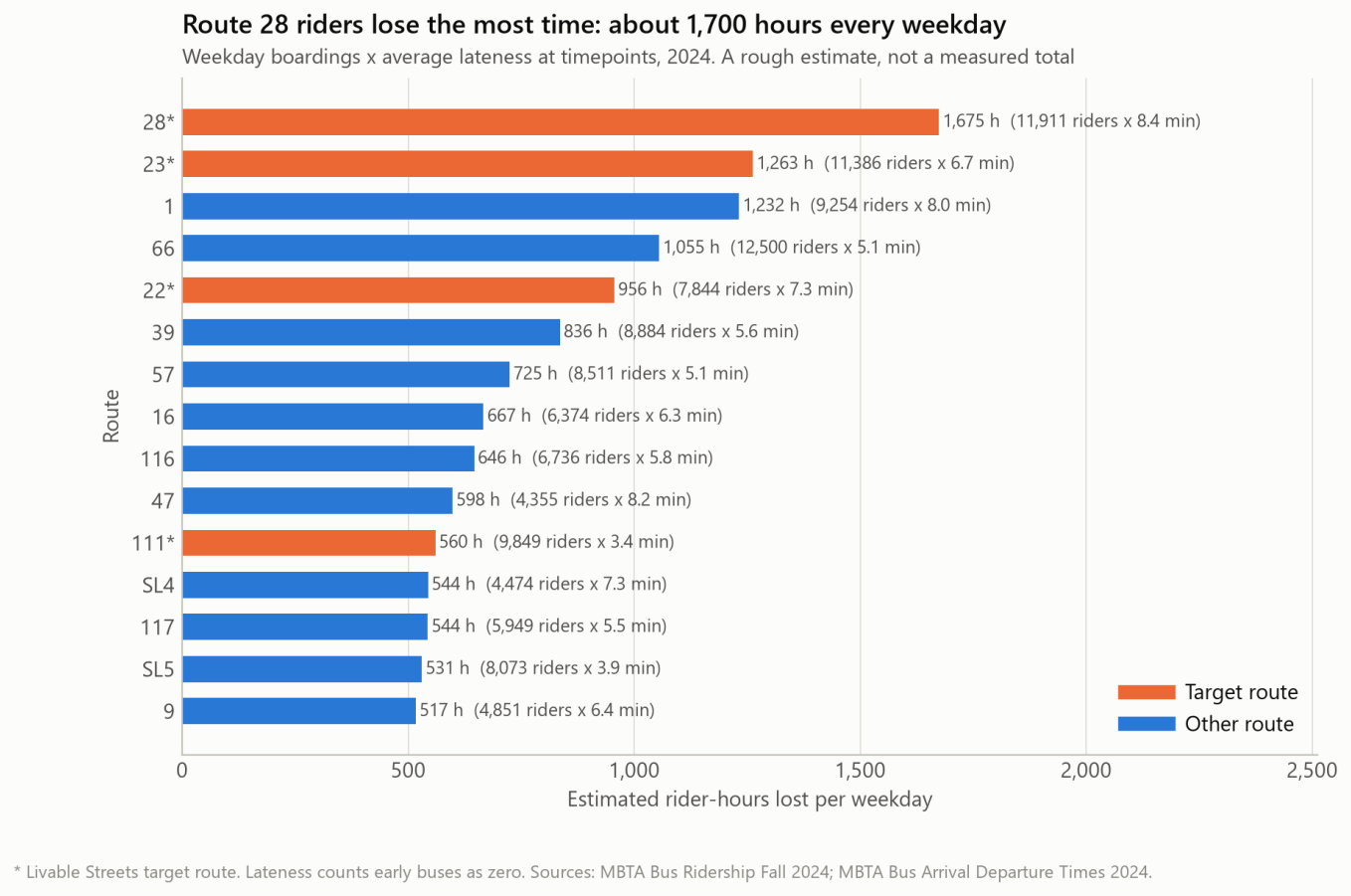
Figure 9. Neighborhood demographics by tract

Verdict: Partly true: on-time rates are about the same across neighborhoods, but buses run somewhat later in minority and lower-income areas.



Figure 10. Rider-hours lost

Verdict: Weakly true: the 15 routes losing the most time account for 45% of lost hours, but they also carry 38% of riders.



* Livable Streets target route. Lateness counts early buses as zero. Sources: MBTA Bus Ridership Fall 2024; MBTA Bus Arrival Departure Times 2024.